

University of Colorado - Boulder
ECEN 4517 - Experiment 2 Report
Garrett Hurd, Christian Lopez

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1 Setup and Initial Measurements

1.1 Nameplate Data

Nameplate data:

Maximum Power (W)	85
Short Circuit Current (A)	5.45
Open Circuit Voltage (V)	22.2
Rated Current (A)	4.95
Rated Voltage (V)	17.2
Series Fuse (A)	20
Maximum System Open Circuit Voltage (V)	600/715
Short Circuit Current (A)	5.4
Open Circuit Voltage (V)	21.1

The above table shows the PV Panel's specifications as listed on the nameplate.

1.2 Initial Daylight Measurements and Inverter Outputs

Pyranometer Value	1072
Time of Day	1:35 PM
Battery Voltage (V)	12.99
Frequency (Hz)	61.7
With Lightbulb - Power (W)	26
With Lightbulb - Voltage (V)	115
With Lightbulb - Current (A)	0.226

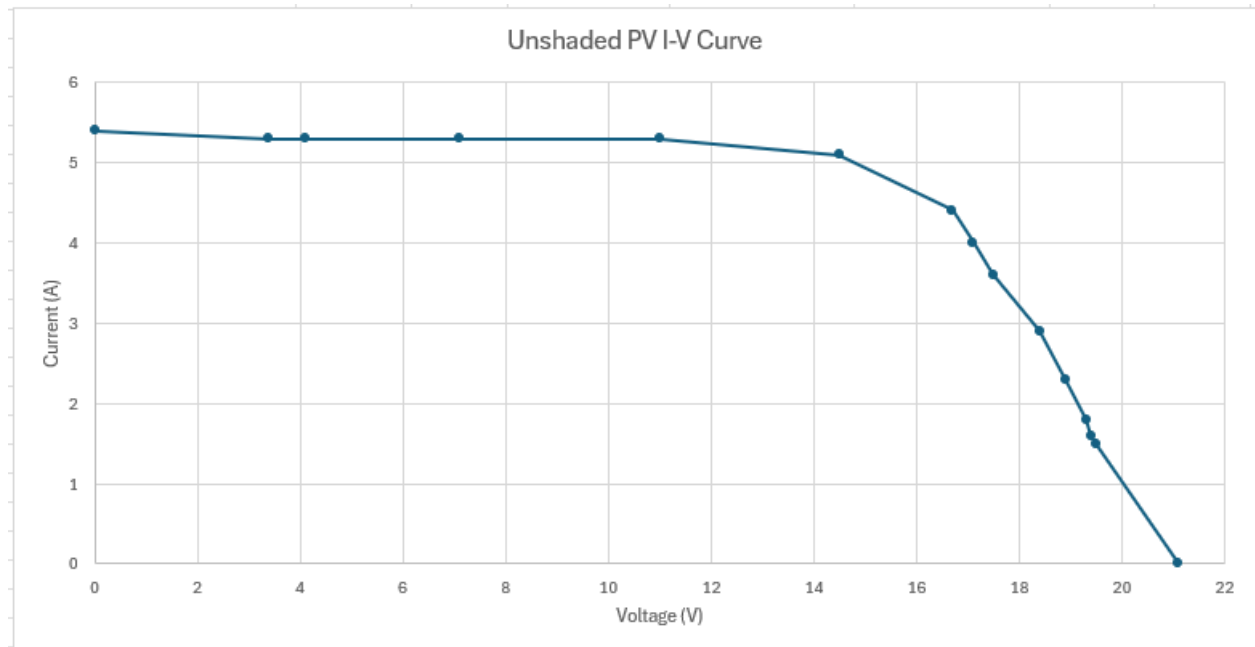
The above table shows the initial measurements taken of AC Inverter, both with and without a load connected.

1.3 Unshaded PV Panel Measurements

Unshaded			
Voltage (cell) (V)	Voltage (Panel) (V)	Current (A)	Power (W)
0.542	19.5	1.5	29.25
0.539	19.4	1.6	31.04
0.536	19.3	1.8	34.74
0.525	18.9	2.3	43.47
0.511	18.4	2.9	53.36
0.486	17.5	3.6	63
0.475	17.1	4	68.4
0.464	16.7	4.4	73.48
0.403	14.5	5.1	73.95
0.306	11	5.3	58.3
0.114	4.1	5.3	21.73
0.197	7.1	5.3	37.63
0.094	3.4	5.3	18.02

Above shows all the PV Panel data collected in order to determine the maximum power point. After adjusting the rheostat, the maximum power point was determined to be 73.95W with a V_{mpp} of 14.5V at 5.1A. These measurements were taken using two multimeters, one configured to read amperage and the other to read voltage. These measurements were recorded at 1:47pm.

1.4 Characterizing the Unshaded PV Panel

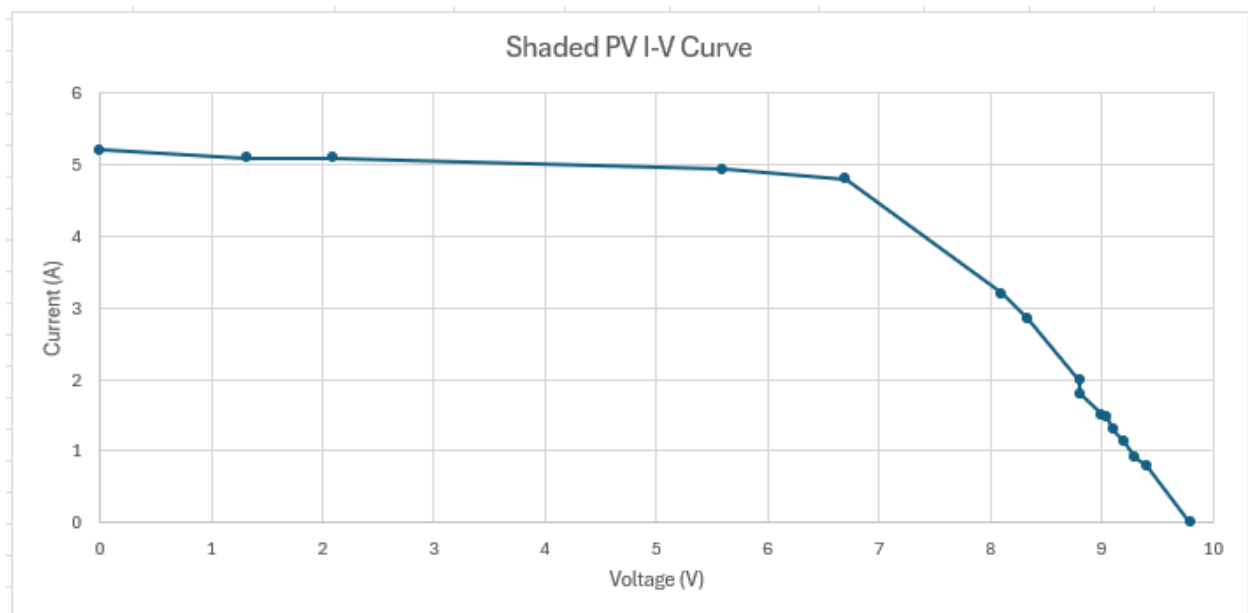


Above is the complete IV curve of the PV Panel. The measured short circuit current was 5.4A and the measured open circuit voltage was 21.1V. The maximum power point remained the same at 73.95W at 14.5V and 5.1A. The curve takes the expected IV curve shape for this device with few to no outliers.

1.5 Characterizing the shaded PV Panel

4 Cells Shaded			
Voltage (cell) (V)	Voltage (panel) (V)	Current (A)	Power (W)
0.261	9.4	0.8	7.52
0.258	9.3	0.9	8.37
0.256	9.2	1.14	10.49
0.253	9.1	1.3	11.83
0.251	9.05	1.47	13.30
0.250	9	1.5	13.50
0.244	8.8	1.8	15.84
0.244	8.8	1.98	17.42
0.232	8.34	2.84	23.69
0.225	8.1	3.2	25.92
0.186	6.7	4.8	32.16
0.156	5.6	4.94	27.66
0.058	2.1	5.1	10.71
0.037	1.32	5.1	6.73

The above table shows the data collected in order to determine the maximum power point with 4 cells of the PV Panel being shaded. The maximum power point was determined to be 32.16W at 6.7V and 4.8A. This data sponsored the creation of the following IV curve shown below. The short circuit current was measured at 5.3A, and the open circuit voltage was measured at 9.8V.



2 Charging the Battery

2.1 Measurements of Charging Battery

Battery Voltage (V)	12.8
Battery Current (A)	5.0
Power (W)	64

The above data was collected while using the PV Panel to charge the battery on the cart. The PV Panel was able to output 64 W of power meaning that it was not operating at its maximum power point.

Assuming the measured data reflects the true maximum power point, then the 64W represents 86.6% of its total power potential.

3 Data Analysis and Modeling

3.1 Efficiency Calculations of PV Panel

After visiting the NREL website, it may be estimated that the direct normal insolation was 1100 W/m². Using this estimation, the efficiency of the panel was calculated to be 10.63% using the equation below.

$$\text{Area of the Panel} \times \text{Direct normal insolation} = P_{in}$$

$$V_{mpp} \times I_{mpp} = P_{out}$$

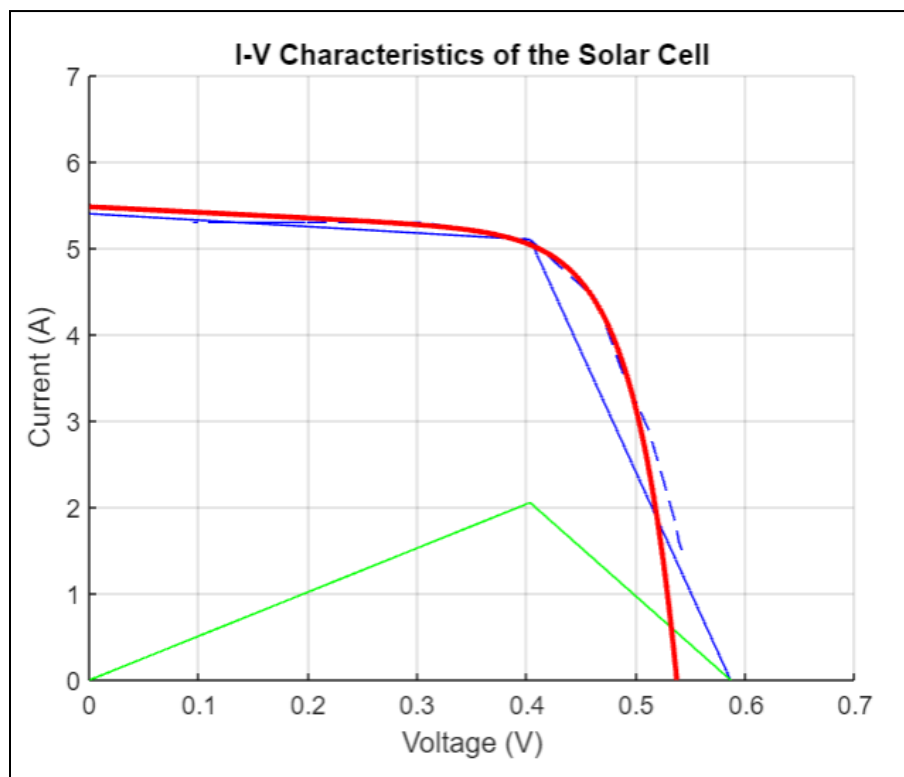
$$\frac{P_{out}}{P_{in}} (100) = \text{Efficiency}$$

The above equations list the power out using the maximum power point voltage and current measured. Compared to the datasheet for the PV Panel, the efficiency calculated was lower than expected. The datasheet describes the efficiency of the panel, at an irradiance of 1000W/m², as 13.4%. The discrepancy seen between the datasheet and the measured efficiency could be linked to the temperature outside on the day this data was collected. However, this is not for certain and more research would be needed in order to determine the root cause.

3.2 Parameter Values for Equivalent Circuit Model

Ki	0.00515
Ido	820×10^{-12}
Rs	0.0114
Rp	1.54

The above table displays the equivalent circuit model parameters found by the MATLAB model.



The above plot shows 4 different plots in different colors and line styles. The solid red line depicts the model I-V curve of the PV panel. The Dashed blue line, partially hidden by the red line, shows the experimentally measured data. The solid blue line represents the Voc, Vmpp, Impp, and Isc values. The green line is the same as the solid blue line but representative of power with Voc x Ioc, Vsc x Isc, and Impp x Vmpp.

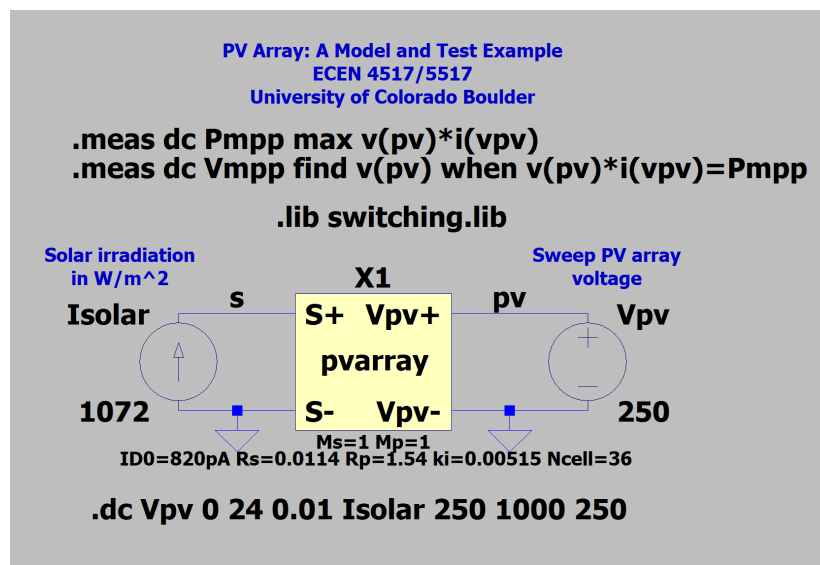
The graph of the experimental data vs the model appears to be very good at modeling our data. However, the open circuit voltage and short circuit current appear to be incorrect. Our model predicts that the open circuit voltage will be about 0.530 V while the experimental data reflects a value of 0.580 V.

Experiment 1		Experiment 2	
Vmpp (V)	17.2	Vmpp (V)	14.5
Impp (A)	4.95	Impp (A)	5.1
Voc (V)	22.2	Voc (V)	21.1
Isc (A)	5.45	Isc (A)	5.4
Pmpp (W)	85.14	Pmpp (W)	73.9

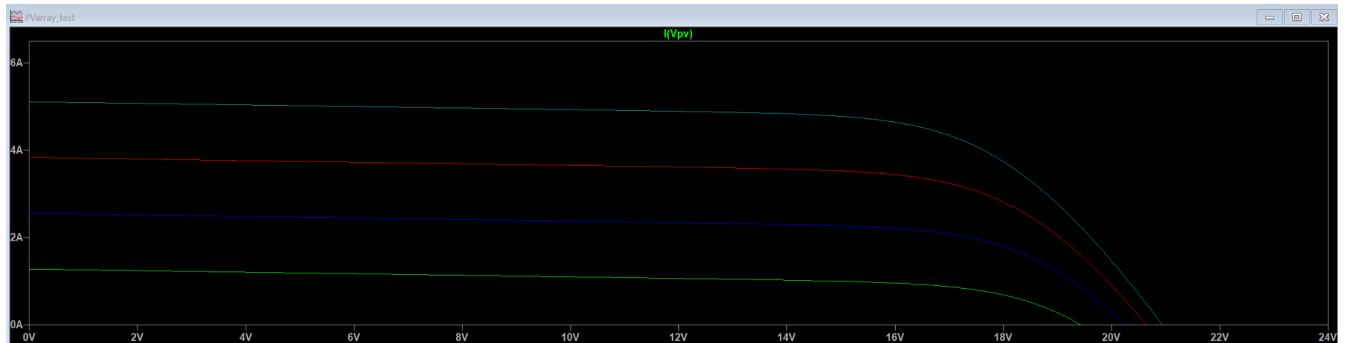
Measurement: pmpp			
step	MAX(v(pv)*i(vpv))	FROM	TO
1	15.292	0	24
2	35.579	0	24
3	55.4603	0	24
4	74.6044	0	24
Measurement: vmpp			
step	v(pv)	at	
1	16.0813	16.0813	
2	16.55	16.55	
3	16.55	16.55	
4	16.41	16.41	

The above table shows the PV panel voltage and current measurements for maximum power point and maximum current and voltage. The left side is our data we got from Experiment 1. The right side is our data from Experiment 2. Our Experiment 2 data appears to be lower than the Experiment 1 data with our Pmpp being 13.3% lower than the Experiment 1 data.

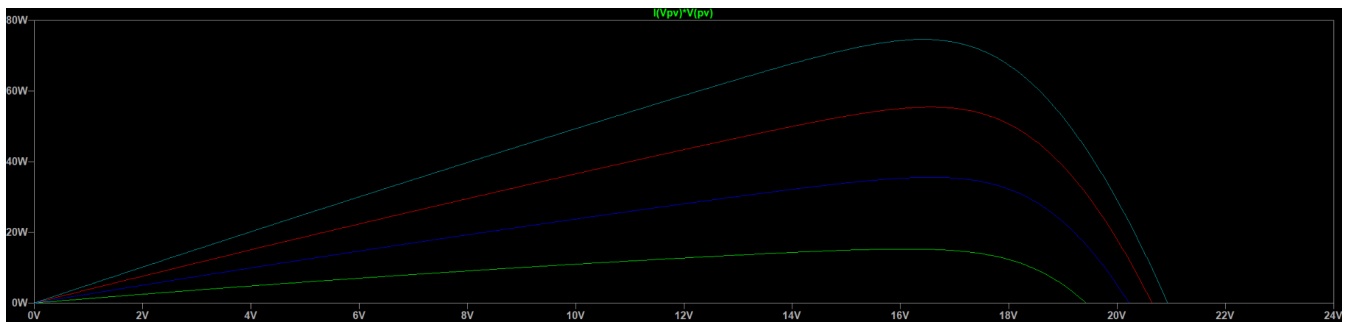
3.3 LTSpice Analysis of Circuit Model Parameters



The two graphics seen above are the LTSpice schematic used to verify our modeled equivalent circuit parameters and the LTSpice output log after running the simulation. It shows a Pmpp of 74.6 W which is only about 1% off from what we experimentally measured.

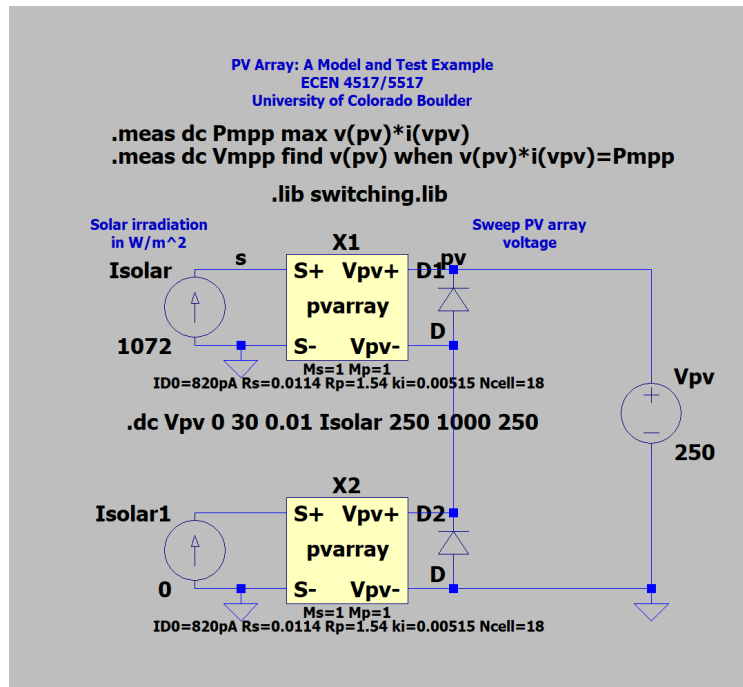


Above is the I-V curve of the PV Panel, showing a Voc around 21 V which was expected for the 1000 W/m² irradiance level.

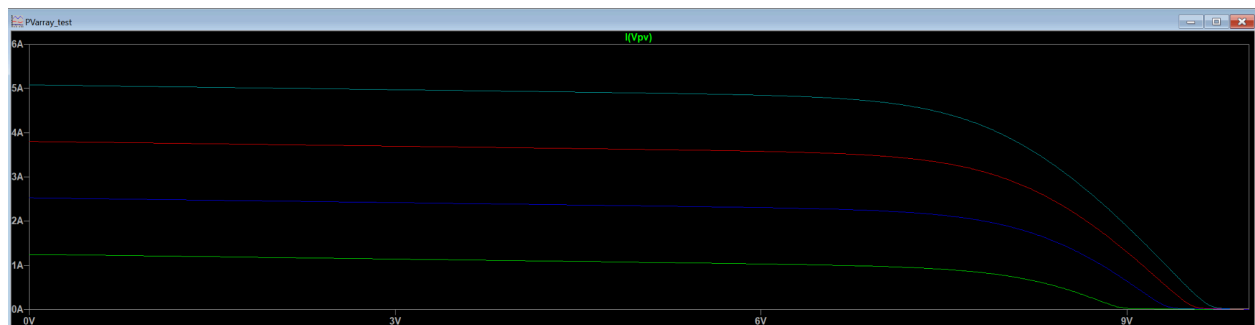


Above is the P-V curve of the PV Panel, showing a Pmpp of about 75 W at 16.4 V.

3.5 PV Panel LTSpice Simulation with Backplane Diodes



Above is the schematic of the shaded PV panel with the two backplane diodes.



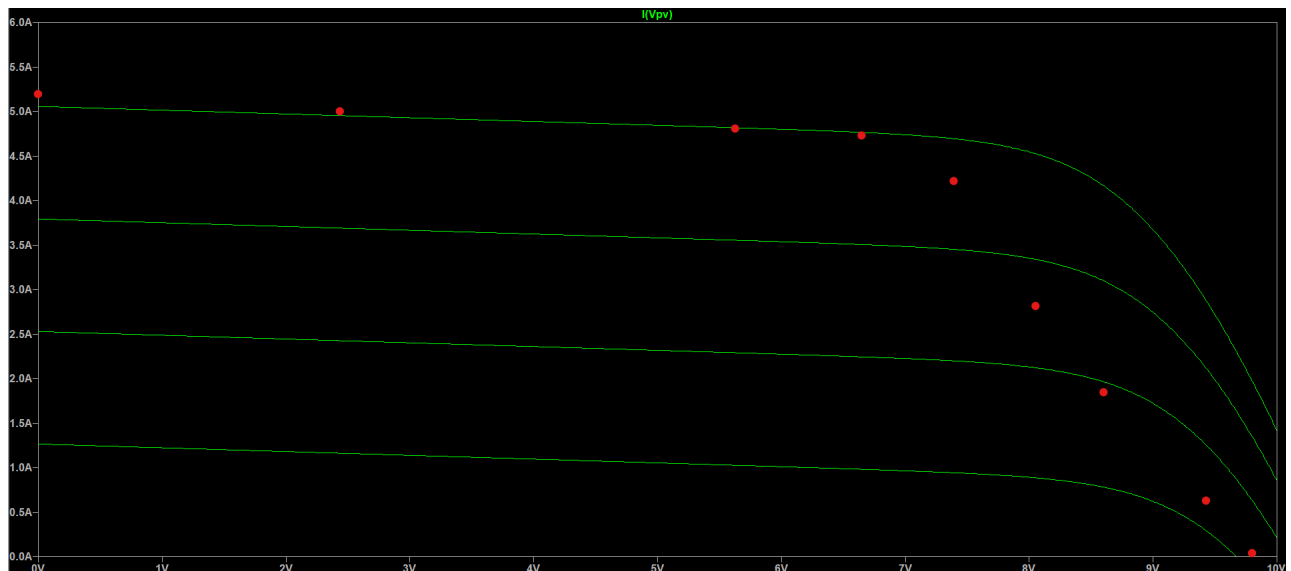
Above is the I-V curve of the shaded PV panel. This curve is very similar to the curve we created from our collected shaded data in section 1.5. Based on our shaded PV panel data we collected, we are confident that the schematic we created and simulated in LTSpice is accurate. The open circuit voltage in the simulation was measured at 9.75 V while the Experiment 2 data listed the open circuit voltage at 9.8 V. The short circuit current was measured experimentally as 5.3 A while the simulation measured it at 5.01 A. The maximum power point was experimentally measured at 32.16 W while the simulation measured it at 33.35 W, as seen in the figure below. At the 32.16 W Pmpp, a Vmpp of 7.4 V with an Impp of 4.50 A. Out of the two diodes, only D2, the bottom diode, conducts any current in this simulation.

Measurement: pmpp		
step	MAX (v (pv) * i (vpv))	FROM
1	6.85868	0
2	15.9622	0
3	24.8436	0
4	33.3503	0

Measurement: vmpp		
step	v (pv)	at
1	7.28	7.28
2	7.48	7.48
3	7.47	7.47
4	7.4	7.4



Above is the P-V curve for the PV Panel with two backplane diodes.



Above is the IV Curve of the shaded PV Panel with the experiment 2 data overlaid on top. A third party program other than LTSpice was used to overlay the data points.

4 Battery Characterization

4.1 Battery State of Charge Calculations

Time of Day	1:35pm	2:03pm
Battery Voltage (V)	12.99	12.56

Based on the above data collected during experiment 2, it may be estimated that the battery was used for 28 minutes and lost 430mV. Using the current draw measurement from section 1.2, the inverter supplied 0.226 A of current with the load lightbulb plugged in. This corresponds to 0.1107 Ah delivered by the battery after accounting for the inverter efficiency of 95%. The below equation shows the exact calculation used.

$$Ah = 0.2261A \times \frac{28 \text{ min}}{60 \text{ min}} (1.05) = 0.1107 Ah$$

Based on table 1 of the Experiment 2 procedure, the ending battery voltage measured was 12.56 V which would correspond to a SOC of roughly 75%. The battery state of charge is not an expected relationship. The SOC level can be estimated as a linear relationship between 75% and 25%, but dramatically changes outside of this range. It is believed that this is to keep the battery in a healthy state, and not allow the battery to drop below 80% of its max charge.

4.2 Battery Operating Point Characteristics

Battery Voltage (V)	Battery Current (A)	Battery Power (W)
12.8	5	64

The above table shows the measurements taken while using the PV Panel to directly charge the battery on the PV cart.

Neglecting losses in the buck converter, the MPPT would increase the voltage and current to the battery. Based off of the Experiment 2 V_{mpp} and I_{mpp} measurements, the P_{mpp} should increase from 73.9 W to 85 W. The 85 W comes directly from the data sheet and what the PV Panel is capable of doing. This would correspond to a power increase of 14% and a current decrease of 4%. This is because the measured P_{mpp} from this lab had a higher current and lower voltage than what the datasheet listed, ultimately providing a lower power point. The voltage, however, would increase from the measured 14.5 V to the datasheet's 17.2 V. This would be an increase of 16%

For an alternative approach for the MPPT, we could build a buck boost converter to further hightein the available power to the battery under shading conditions. Another approach would be to use a more adaptable MPPT algorithm that is trained to adjust to shading and non shading conditions.

4.3 Preliminary DC-DC Converter Specifications

- a) The converter input voltage of 3.4 V to 19.4V and a current of 1.5 A to 5.3 A
 - i) These are the experimentally gathered results of the output of the PV Panel.
- b) The converter output voltage is 10.5V to 13 V and a current of 0A to 5.3A
 - i) These are the experimentally gathered results of the battery's SOC Voltage and current.
- c) The converter switching frequency is going to be 100k Hz, we used this value in the prelab for experiment 3 to design the inductor.
- d) The efficiency of the buck converter at $V_{in} = 17V$ and $V_{out} = 12V$ is 70%. $\frac{V_{out}}{V_{in}}$ is the M(D) equation for an ideal buck converter.